

MCQ:

20x1=20

1. String s= "Hello"; System.out.println(s.substring(1, s.lastIndexOf('l')+1).toUpperCase().endsWith("L"));

- a) Hell b) ELL
c) true d) false

2. System.out.println(Math.pow(2, Math.pow(49, 1/2)))

- a) 128 b) 128.0
c) 1.0 d) 2.0

3. String s= "Hello"; int a=10; double b=20.0;

System.out.println(s+a+b);

- a) Hello1020.0 b) Hello1020
c) Hello30.0 d) Hello10.020.0

4) Find output: int i;

for(i=1; i <= 10; i+=4); System.out.println(i);

- a) 1 b) 9
c) 13 d) 41

5) String a[]={ "JAVA", "JavaScript", "Python", "C++", "Solidity", "Go"};

int k= a[a[3].length()+1].toUpperCase().compareTo("UNIVERSITY");

- a) 2 b) -2
c) error d) -3

6) Which one is in correct order of precedence

- a) ++ ! / - > b) ! / ++ - >
c) / - ++ ! > d) / + ! ++ -

7. int k=1, n=sc.nextInt();

```
do{
    //some code
}while(k< n);
```

This loop can run a minimum and maximum of ____ and ____ times

1. 1 and n 2. 1 and infinite
3. 0 and infinite 4. 0 and n

8. int a=10,b=20; Here a and b occupy 4 bytes each. int s=a+b; how much space would s occupy?

- a) 8 bytes b) 4 bytes
- c) 12 bytes d) 13 bytes

9. Escape Sequence for new line and carriage return is

- a) \n and \b b) \n and \c
- c) \n and \t d) \n and \r

10. if(k== (n=sc.nextInt())) checks if k is equal to n. This statement is _____

- a) true b) compilation error
- b) false d) runtime error

11. arrange the data types in ascending order : int, char, double, short

- a) char, short, int, double b) short, char, int, double
- c) double, int, short, char d) int, char, short, double

```
12. class Fruit{  
    int a;  
    Fruit(Fruit ob){  
        a=ob.a;  
    }  
}
```

It's a

- a) default constructor b) Parameterized Constructor
- c) Copy Constructor d) Logical Error

13) "this" keyword is used to refer to

- a) current instance/object of class b) current class
- b) parameterized object d) member of parameterized object

```
14) class Apple{  
    static int[] BubbleSort( int a[] ){  
        a[0]=a[a.length-1];  
        return a;  
    }  
    public static void main(String args[]){  
        int a[]=BubbleSort(new int[]{1, 2, 3, 4, 5});  
        System.out.println(a[0]+""+a[a.length-1]);  
    }  
}
```

- ```
15) import java.util.Scanner;
class Main{
```

```
18) class SaptaparnaTheGreat{
 static int yo(int x){
 return x*x;
 }

 public static void main(String args[]){
 System.out.println(yo(yo(yo(yo(2)))));
 }
}
```

- a) Error                                      b) 256  
c) 65536                                      d) 128

19. char ch = "";

System.out.println(ch);

- a) Error (Empty Character Literal)                      b) Prints Blank  
c) \u0000                                      d) NULL

20. int r= "Akuresh".compareTo("Akureshii");

System.out.println(r);

- a) 5                                              b) -1  
b) -2                                              d) 0

**SAQ. Answer the following (Showing dry run is compulsory for snippets)**

**10x2=20**

```
1. class Apple {
 public static void main(String[] args) {
 int a = 3, b = 7, result = 1;

 switch (a + b) {
 case 8:
 result += a;
 System.out.println("Case 8: result = " + result);
 case 10:
 result *= b;
 System.out.println("Case 10: result = " + result);
 switch (b % 3) {
 case 0:
 result -= b;
 System.out.println("Nested Case 0: result = " + result);
 case 1:
 result += a * b;
 System.out.println("Nested Case 1: result = " + result);
 case 2:
 result /= a;
 System.out.println("Nested Case 2: result = " + result);
 }
 case 12:
 result += b - a;
 System.out.println("Case 12: result = " + result);
 default:
 result -= 5;
 System.out.println("Default: result = " + result);
 }

 System.out.println("Final result = " + result);
 }
}
```

```

2. for(int i=1; i<=5; i++){
 for(int j=1; j<=5; j++){
 if(i+j>=6){
 System.out.print(j-(5-i)+ " ");
 }else{
 System.out.print(" ");
 }
 }
 System.out.println();
}

```

Draw the number pattern generated this loop.

3. Write the following expression in JAVA.

$$k = \sqrt{|ax| + a^x + x^a}$$

```

4. int x= 5, y=10;
 x -= ++x + y-- + --y + x-- + x + 4%8;
 x=? y=?

```

5. Find output and the number of times the loop executes

```

long n=1234567890, c=0;

while(n>0){

 long d= n%10;

 System.out.println(d);

 n/=Math.pow(10,c++);

}

```

6. find output and the number times the loop executes

```

int p=1;
do{
 p*=2;
 System.out.println(p);
}while(p!=32);

```

7. int a,b,c,d;

a=b=c=d=4;

a=b=c=3;

a=b=2;

a=1;

System.out.print("a="+a+" b="+b+" c="+c+" d="+d);

8. int a; float f; char ch; short s;

Assume an arithmetic expression is formed using the above variables and the result is stored in k. find data type of k.

9. Find the errors, write the type of errors you can see, and correct them (if possible)

```
class apple juice{
 public static void main{
 int a= 6%3;
 int b= a/a;
 System.out.println("Sum of two numbers="+b);
 }
}
```

```
10. for (int i = 1; i <= 3; i++) {
 int j = 1;
 while (j <= 2) {
 System.out.print(i + " " + j + " "+ (i+j));
 j++;
 }
}
```

Find output and number of times the loop executes.

### Section B

Write programs in  and write VDTs

Solve any four:

15x4=60

1. Write a program to take distance and AC information ( 1 if AC is on and 0 if AC is off) from user calculate taxi fare according to tariff given below.

class name: Taxi

Data Members:

|          |                                    |
|----------|------------------------------------|
| int d    | to store distance                  |
| int fare | to store fare                      |
| int ac   | to store 1 or 0 for ac information |

Member methods:

Taxi() to initialize data members to default values

accept() to take distance from user

calculate() to calculate fare

display() to display distance and fare

| Distance    | Fare per kilometre      |
|-------------|-------------------------|
| upto 5 km   | Rs. 150                 |
| Next 20 km  | Rs. 22.50 per kilometre |
| Above 25 km | Rs. 30 per kilometre    |

The taxi charges Rs. 100 extra if the passenger wants the AC to be turned on.

2. Write a menu driven program to do the following:

(a) Print the number pattern using nested loop:

```
1
2 1
3 2 1
4 3 2 1
5 4 3 2 1
```

(b) Print the following number pyramid using nested loop:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

3. Write a program to create a matrix of N x N size, fill it with Pell Series in Row major order.

The **Pell series** is a sequence of numbers defined by the recurrence relation:

$P(n) = 2 \times P(n-1) + P(n-2)$  or ( a, b , a+2b .....)

2) Pell Sequence: 0,1,2,5,12,29,70,169,408,985,...till N

4. Take a number and check if it's a duck number or not.

Duck number is a number such that it must contain 0 (one or more) but should not start with 0

5. Take a String from user and generate all possible substrings, also count the occurrences of another substring s2 given by user.

6. Make a function BubbleSort(int a[]) to take an array and return the sorted array (ascending order)

Write the driver code to take an array from user and sort it using BubbleSort() function